

Program: Diploma in Electrical Engineering & Electrical Vehicle Technology	
Course Code: 3411	Course Title: Electric Vehicle Batteries & Charging Systems.
Semester: 3	Credits: 4
Course Category: Program Core	
Periods per week: 3 (L:2 T:1 P:0)	Periods per semester: 45

Course Objectives:

- To understand the different types of energy storage systems.
- To study the battery characteristics & parameters.
- To model the types of batteries
- To know the concepts of battery management system and design the battery.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Basic knowledge about storage batteries	2411	Introduction to Electric Vehicles & Elementary Theory	2
Basic Concept of Electro Chemistry	1004	Applied Chemistry	2

Course outcomes:

On Completion of this course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Outline the features of batteries used in vehicles.	10	Understanding
CO2	Discuss the construction and working of batteries used in vehicles.	11	Understanding
CO3	Understand the charging systems of batteries.	12	Applying
CO4	Identify the EV charging technologies.	10	Understanding
	Series Test	2	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Outline the features of batteries used in vehicles		
M1.01	Explain the various parameters of batteries used in vehicles.	3	Understanding
M1.02	Identify the various electrical components in EVs	2	Understanding
M1.03	Significance of earthing in EV systems	2	Understanding
M1.04	Identify the methods of earthing in EV systems	3	Understanding
Contents: Battery parameters: Battery parameters: Physical Dimensions, Voltage and current rating, Capacity, and power, C Rate, Battery Efficiency, Energy Density, Power Density, State of charge (SOC), Depth of discharge (DOD), State of Health (SOH), Operating Temperature, Lifetime, Battery Management System(BMS) Various Electrical components/systems in Automobile- Main components of EV subsystems– Electric Propulsion unit, Energy source subsystem, auxiliary subsystem, Drive train in EV: Basic Architecture of Electric Drive Train, Earth return system, Functions of earthing systems in EV systems- Types of earthing systems- TT earthing, TN earthing, GFCI protection for EV Charging stations.			
CO2	Discuss the construction and working of batteries used in vehicles.		
M2.01	Identify the various types of batteries used in EVs	3	Understanding

M2.02	Describe the construction and working of Lithium based batteries.	4	Understanding
M2.03	Comparison of batteries used in EV systems	2	Understanding
M2.04	Importance of ultra-capacitors in EV systems.	2	Understanding
	Series Test I	1	
Contents: Electrochemical batteries: Lead Acid Batteries – Nickel based batteries – Lithium based batteries - Fuel Cell. Construction and working of lithium-based batteries- Advantages of Lithium batteries. Comparison of batteries with respect to specific energy, specific power, cycle life, cost. Brief introduction of Ultra Capacitors- Importance of ultracapacitors in hybridization of energy storage devices- Advantages			
CO3	Understand the charging systems of EV batteries.		
M3.01	List out the various charging infrastructure for EVs	3	Understanding
M3.02	Explain the types of charging infrastructure.	3	Understanding
M3.03	Explain the conventional charging methods of EVs and HEVs.	3	Applying
M3.04	Outline the concept of wireless charging, e-mobility and electrification.	3	Understanding
Contents: Charging Infrastructure: Domestic Charging Infrastructure, Public Charging Infrastructure, Normal Charging Station, Occasional Charging Station, Fast Charging Station, Battery Swapping Station, Move-and-charge zone. Public charging infrastructure - general requirements (operational and technical). Charging of EV and HEV, AC charging, DC Charging, Battery swapping, Smart charging, Wireless charging. E-mobility and electrification - scope and challenges			
CO4	Identify the EVs charging technologies.		
M4.01	Identify the different charging technologies for EVs.	2	Understanding
M4.02	Explain energy management strategies used in EVs and HEVs.	3	Understanding

M4.03	Outline the concept of wireless power transfer. (WPT).	2	Understanding
M4.04	Explain the various protocols used for communication	3	, Understanding
	Series Test II	1	

Contents:

EV Charging Technologies: EV charging system - block diagram- Classification of different charging technology for EV charging station, introduction to Grid-to-Vehicle, Vehicle to Grid (V2G) or Vehicle to Buildings (V2B) or Vehicle to Home (V2H) operations, bi-directional EV charging systems.

Energy management strategies used in hybrid and electric vehicle-Connectors in EV - types - requirements - input - output - safety - CCS - Bharat EV AC- Bharat EV DC.

Wireless power transfer (WPT) technique for EV charging.

Protocols and communication: Specification of open charge point protocol (OCCP 1.6/2.0) - Bharat DC001 & AC001 Charger specification-CAN communication

Text/Reference:

T/R	BookTitle/Author
R1	James Larminie Oxford Brookes University, Oxford, UK John Lowry Acenti DesignsLtd., UK, Electric Vehicle Technology Explained
R2	C.C Chan, K.T Chau: Modern Electric Vehicle Technology, Oxford University PressInc., New York 2001
R3	Iqbal Hussein, Electric and Hybrid Vehicles: Design Fundamentals, CRC Press, 2003
R4	Mehrdad Ehsani, Yimi Gao, Sebastian E. Gay, Ali Emadi, Modern Electric, HybridElectric and Fuel Cell Vehicles: Fundamentals, Theory and Design, CRC Press, 2004
R5	James Larminie, John Lowry, Electric Vehicle Technology Explained, Wiley, 2003

Online Resources:

Sl.No	Website Link
1	https://www.haikai_energy.com
2	https://bilitielectric.com
3	https://www.spglobal.com
4	https://www.metisengineering.com
5	https://nptel.ac.in/courses/108/103/108103009/
6	https://onlinecourses.nptel.ac.in/noc20_ee99/preview